AVDASI 3

(CADE 30007)
Gas Turbine Propulsion

Week 20: Lecture 7

Turbines

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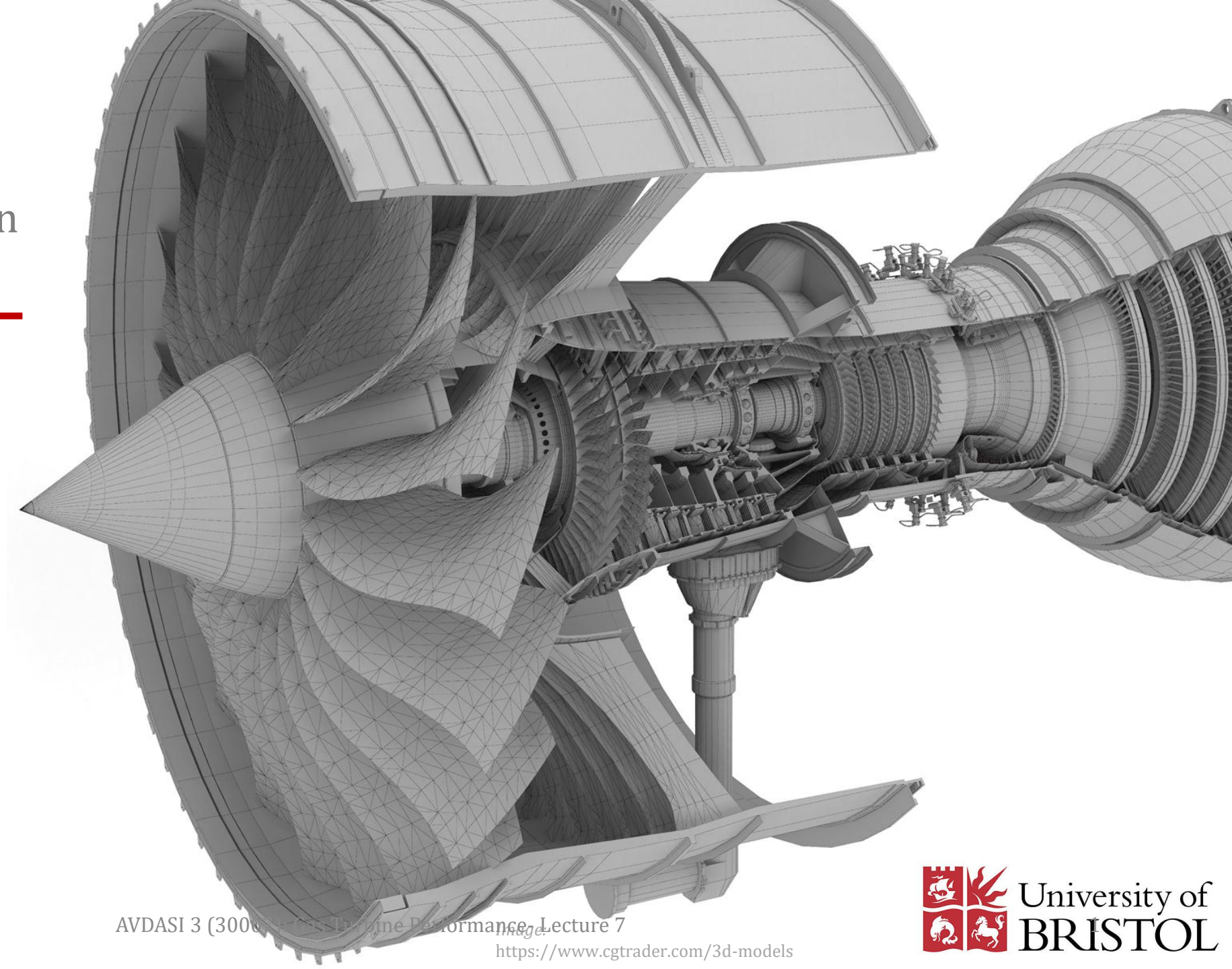

Image: https://www.cgtrader.com/3d-models

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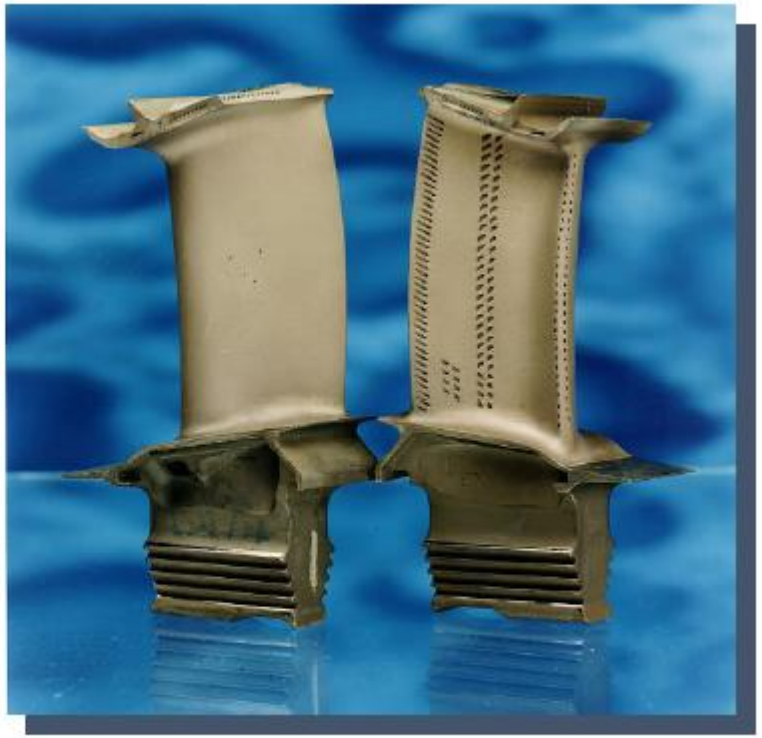
Learning Objectives for Lecture 8

- Objective: to describe the workings of the Turbine

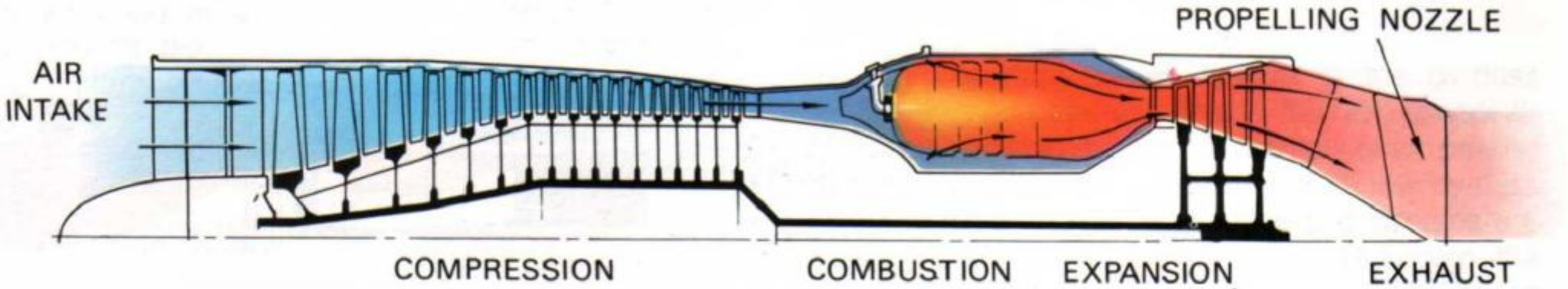
Nozzle Guide Vanes

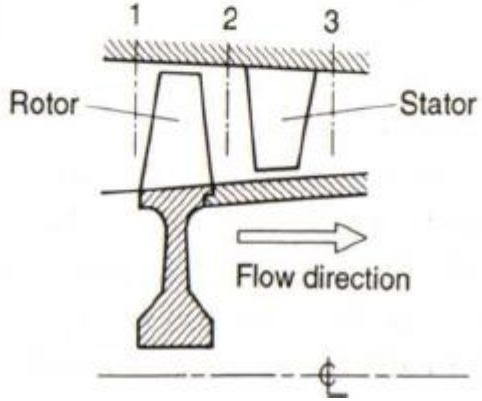


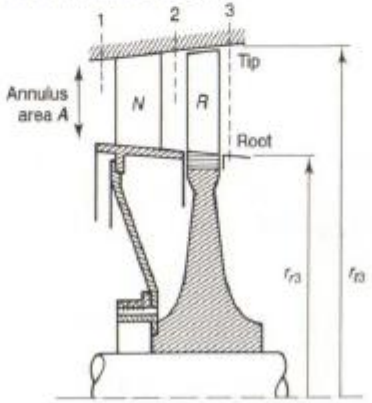
Turbine Blades

Turbomachinery: Introduction

- The **compressor** raises the pressure of the air before combustion
- The **turbine** extracts work from the hot high pressure combustion products to drive the compressor







Each stage consists of a row of rotor blades followed by a row of stator blades
Often preceded by an inlet guide vane

Each stage consists of a row of Nozzle Guide Vanes which direct the gas onto the rotor blade

Turbines: Overall Principles

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- Turbines provide the power to drive the Compressors, Accessories and Power Turbines
- Energy is extracted from the hot gases released from the Combustion Chamber and expanding them to lower Temperatures & Pressures
- Cycle calculations define an overall temperature drop & pressure drop at a given level of isentropic efficiency

Turbines: Overall Principles

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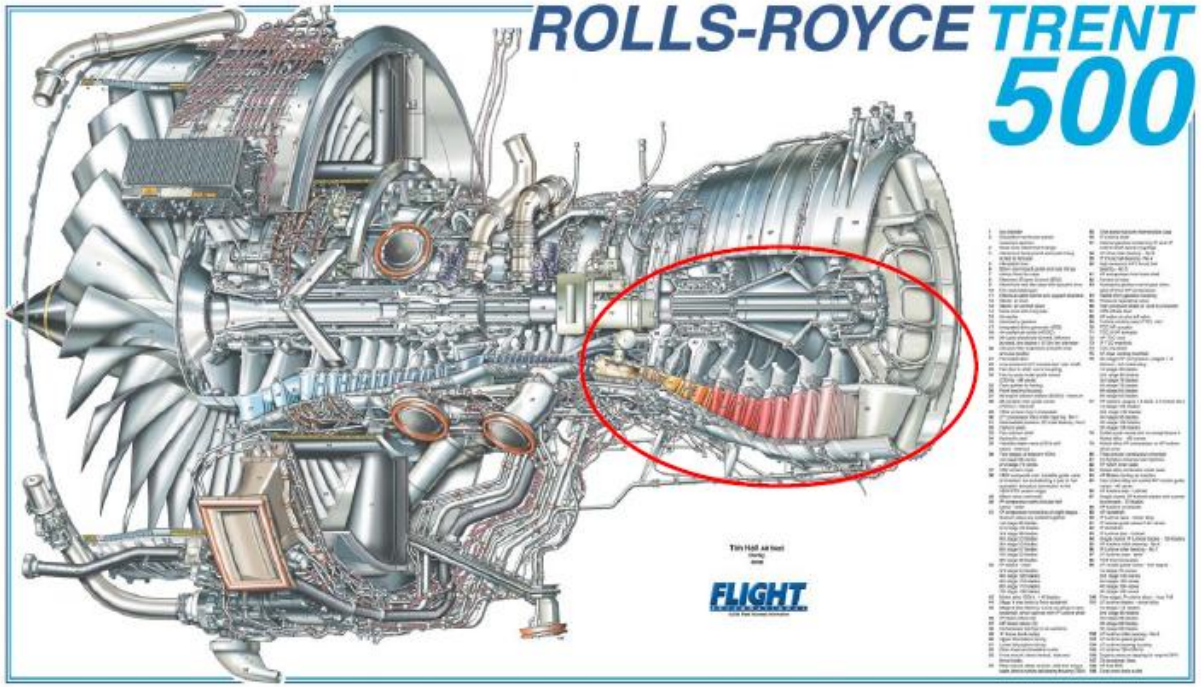
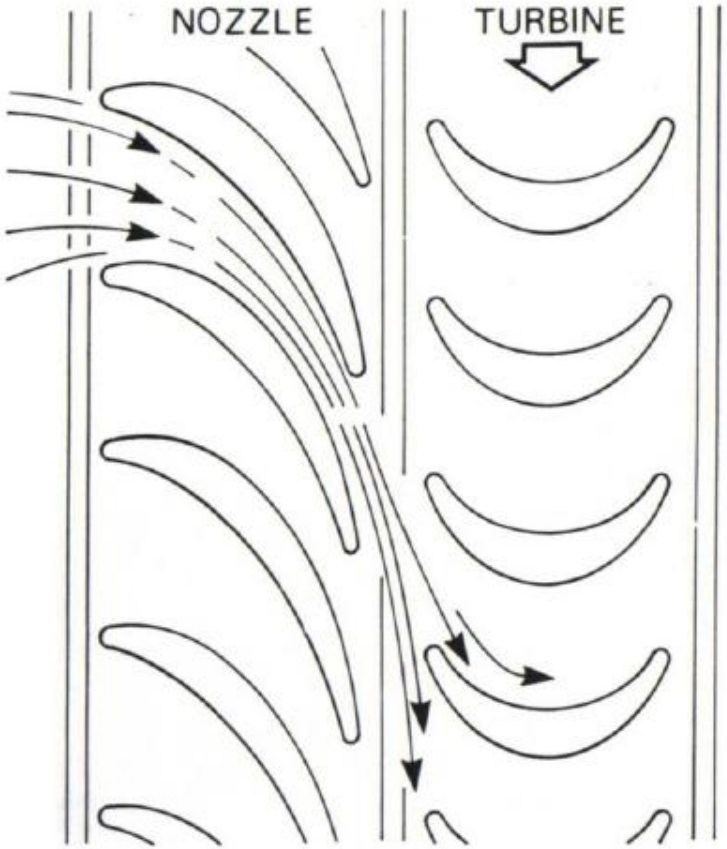
- The rotational speed must be equal to that of the compressor, which is on the same shaft



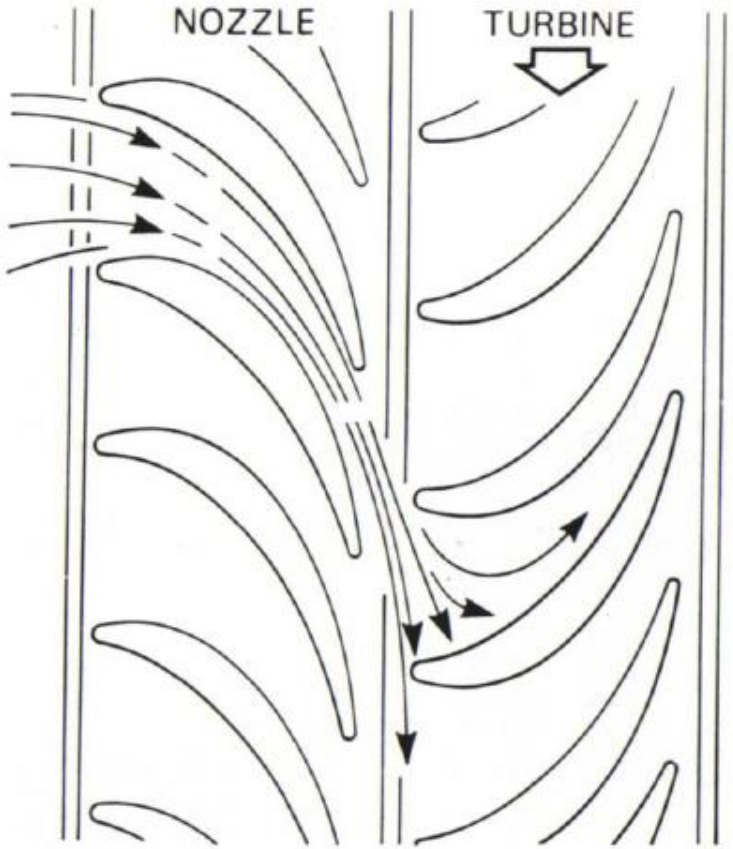
Trent 500 Civil Turbofan (A340)
Single stage HP Turbine (HPC & Accessories)
Single stage IP Turbine (IPC)
5 Stage LP Turbine (Fan)

Multi-shaft arrangements are common to optimise rotational speeds!

Comparison of a Pure Impulse & an Impulse/Reaction Turbine

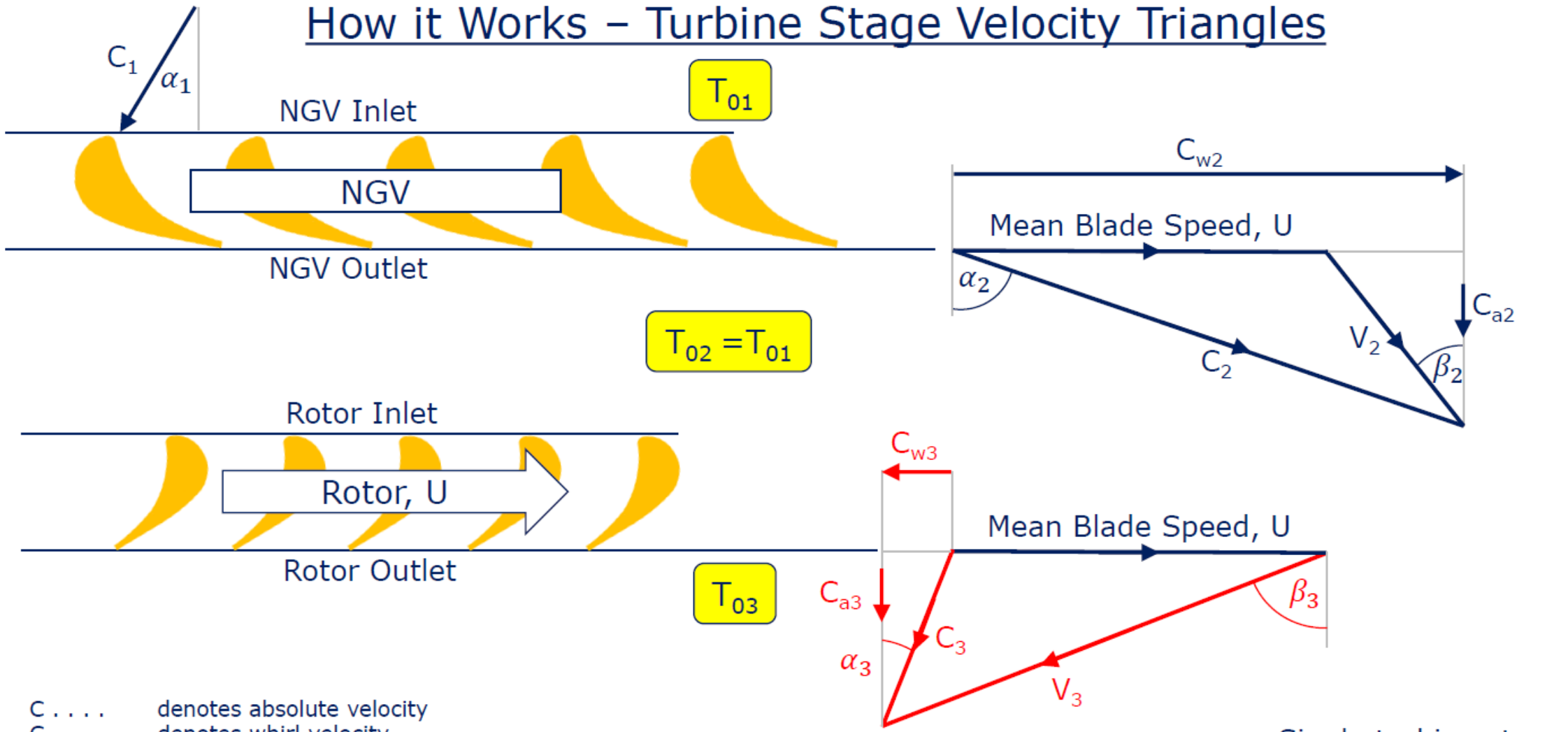


Turbine driven by the impulse of the gas flow only



Turbine driven by the impulse of the gas flow and its subsequent reaction as it accelerates through the converging blade passage

Turbine Velocity triangles



C . . . . denotes absolute velocity
C_w . . . denotes whirl velocity
C_a . . . denotes axial velocity
V . . . . denotes relative velocity (to rotor)

Single turbine stage at **MEAN** radius

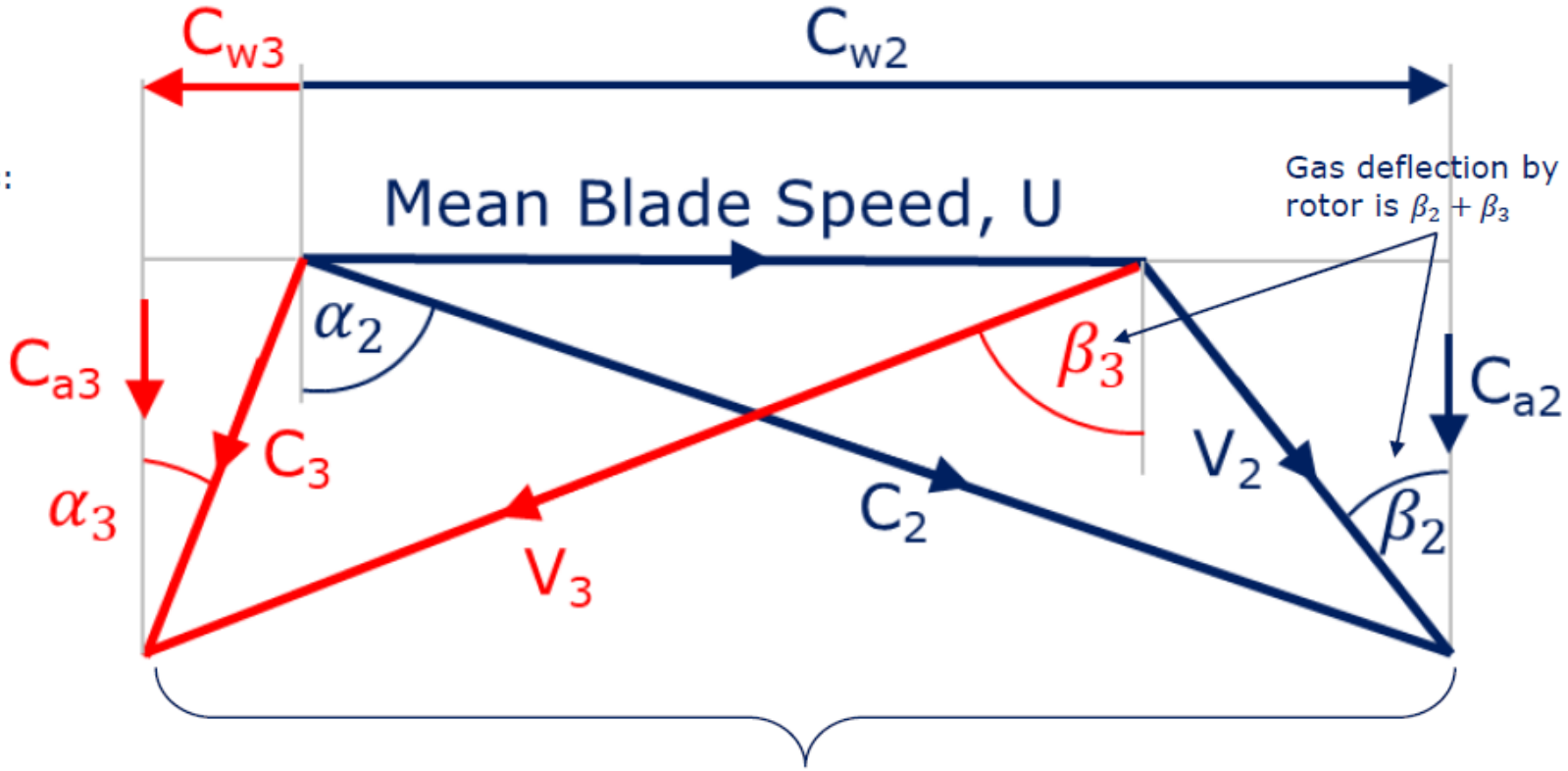
Turbine Velocity triangles

Rotor Inlet
Rotor Outlet

Simplifying assumptions:

$C_{a3}=C_{a2}=C_a$

$C_3=C_1$ and $\alpha_3 = \alpha_1$

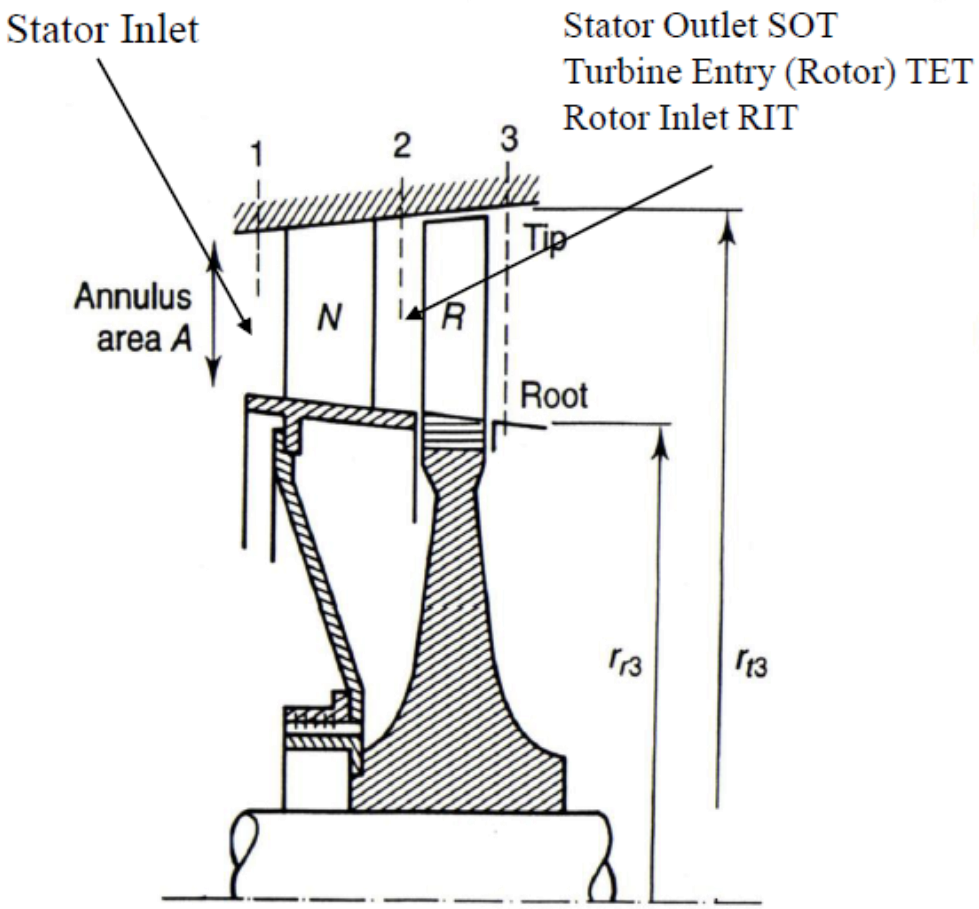


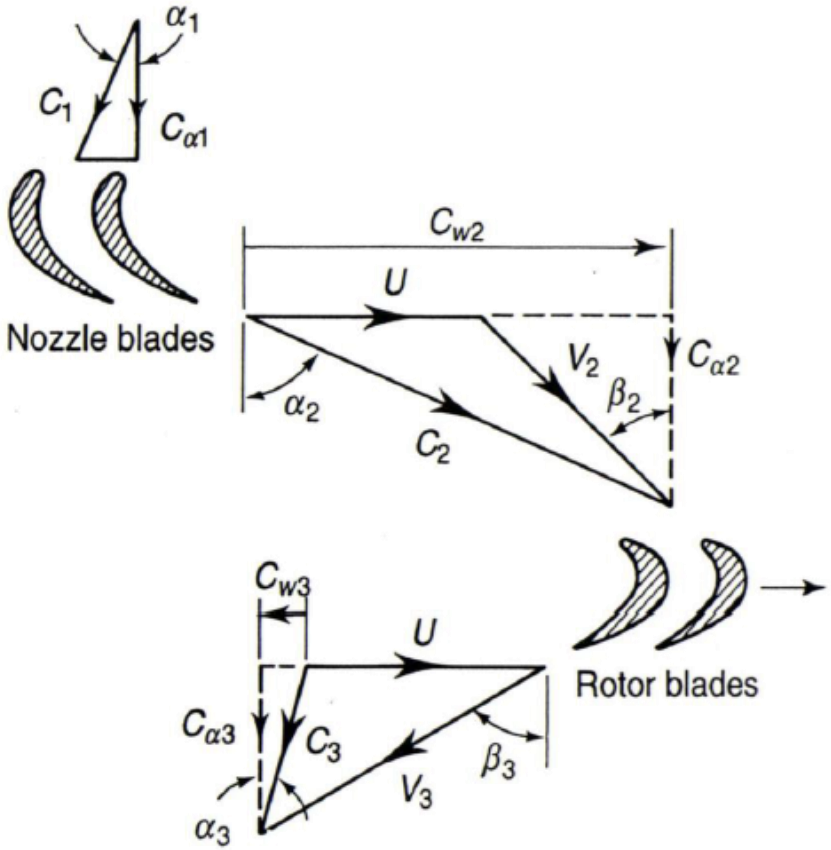
Total Whirl Velocity Change
(proportional to turbine power)

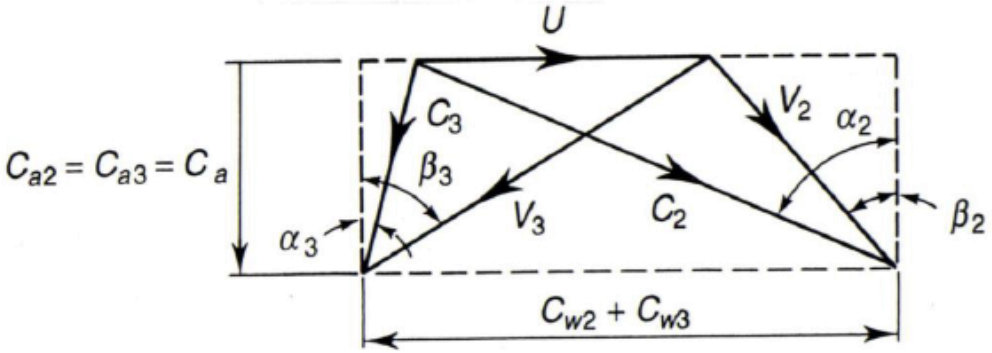
Note that gas is expanded (i.e. accelerated) in NGV from C_1 to C_2 and accelerated in blade passage from V_2 to V_3.

Unlike compressor, vanes and blades in turbine provide big angular gas deflections, possible because of favourable pressure gradient.

Axial Flow Turbine Stage

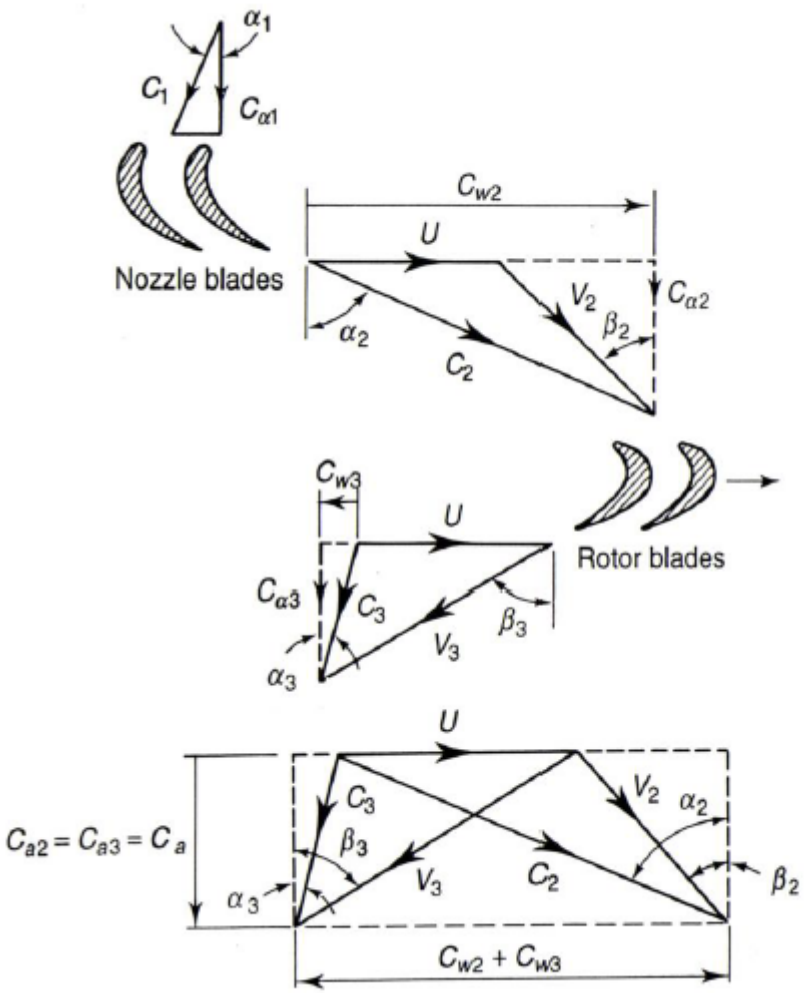






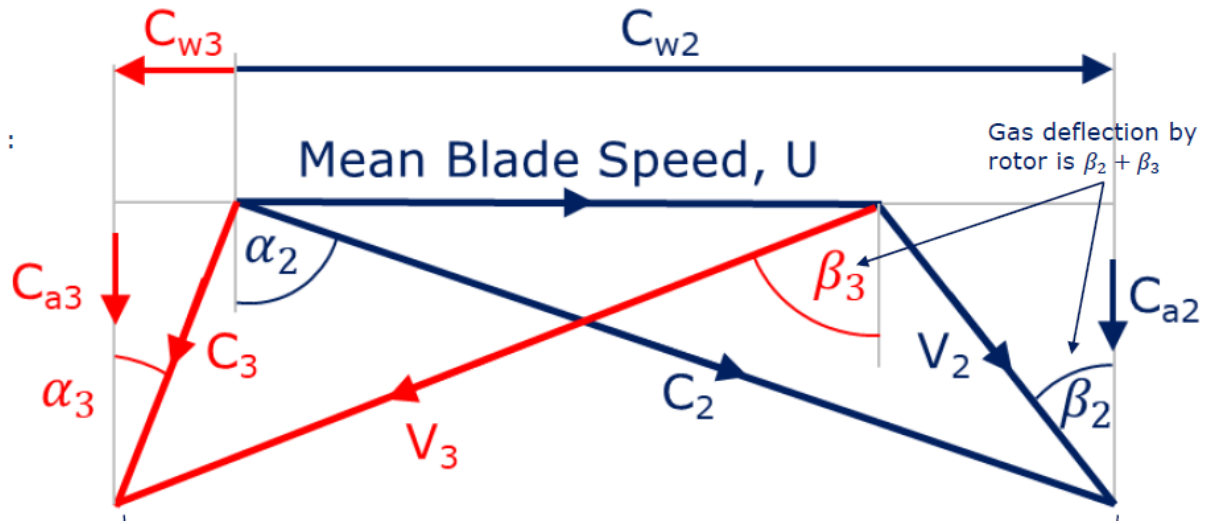
Deflection of flow in rotor = $\beta_2 + \beta_3$

Axial Flow Turbine Stage



	NGV	ROTOR
Total Temperature	Constant	Decrease
Enthalpy	Constant	Decrease
Static Temperature	Decrease	Decrease
Total Pressure	Small Decrease	Decrease
Static Pressure	Decrease	Decrease
Relative Velocity	N/A	Increase
Absolute Velocity	Increase	Decrease
Density	Decrease	Decrease

Axial Flow Turbine Stage



Assume constant axial velocity

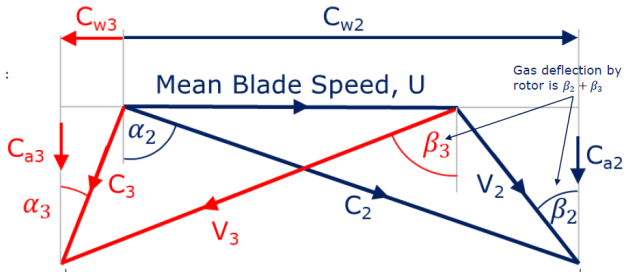
$$C_{a1} = Ca_2 = Ca_3 = Ca$$

From velocity diagrams:

$$\frac{U}{C_a} = (\tan\alpha_2 - \tan\beta_2) = (\tan\beta_3 - \tan\alpha_3)$$

Axial Flow Turbine Stage

Stage Work per unit mass flow $= U \cdot (Change\ in\ whirl\ velocity\ inlet\ to\ outlet)$



$$= U \cdot (Cw_2 + Cw_3) \qquad Note\ sign\ !$$

$$= U \cdot C_a \cdot (\tan\alpha_2 + \tan\alpha_3)$$

$$\Delta h_{os} = C_p \cdot \Delta T_{os} = U \cdot C_a \cdot (\tan\beta_2 + \tan\beta_3)$$

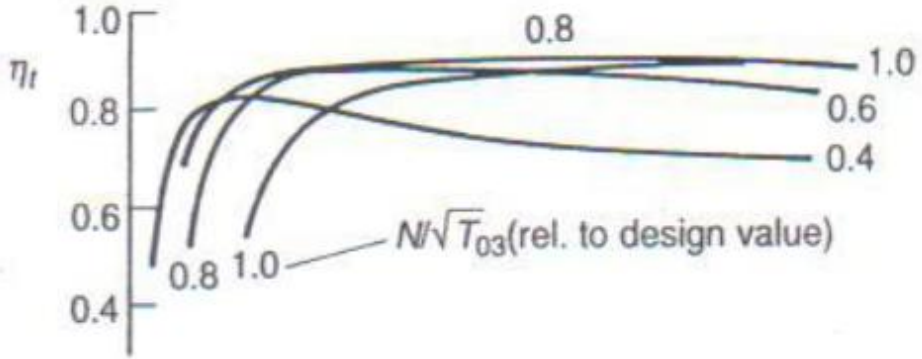
Where ΔT_{os} = Stagnation temperature drop in the stage

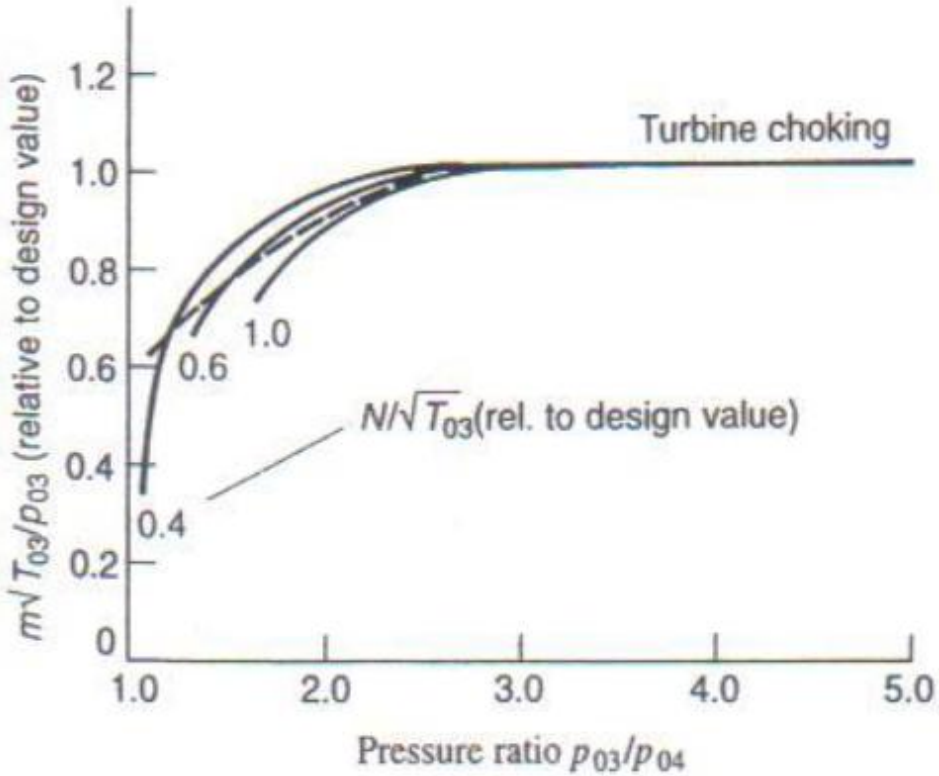
When inlet & outlet velocities are the same

$$C_1 = C_3 \ \ and\ hence\ \Delta T_{os} = \Delta T_s$$

Stagnation Pressure ratio, $\frac{P_{01}}{P_{03}}$ can be found from $\Delta T_{os} = \eta_s T_{01}\left[1 - \left(\frac{1}{p_{01}/p_{03}}\right)^{\frac{\gamma-1}{\gamma}}\right]$

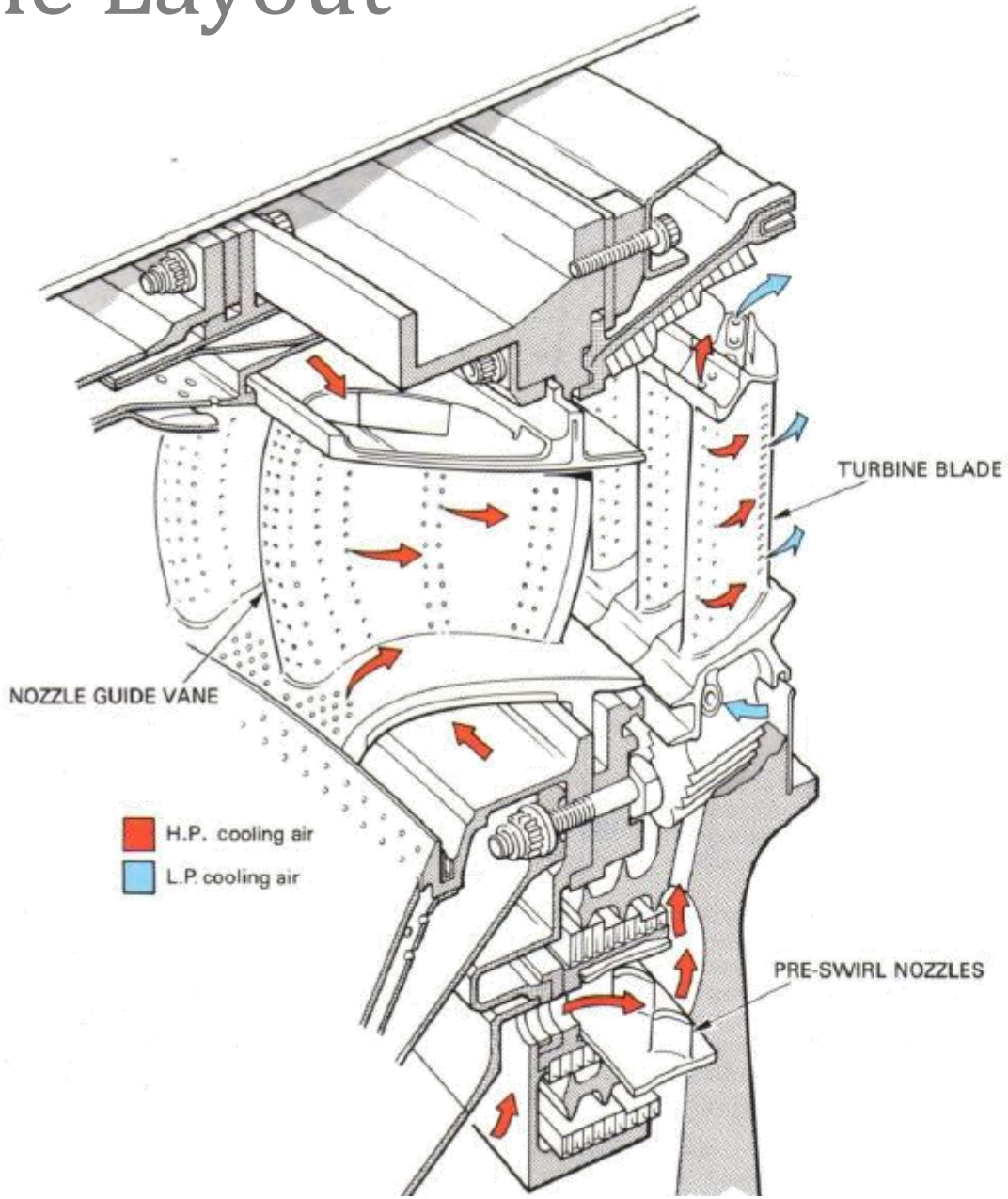
Turbine Characteristics

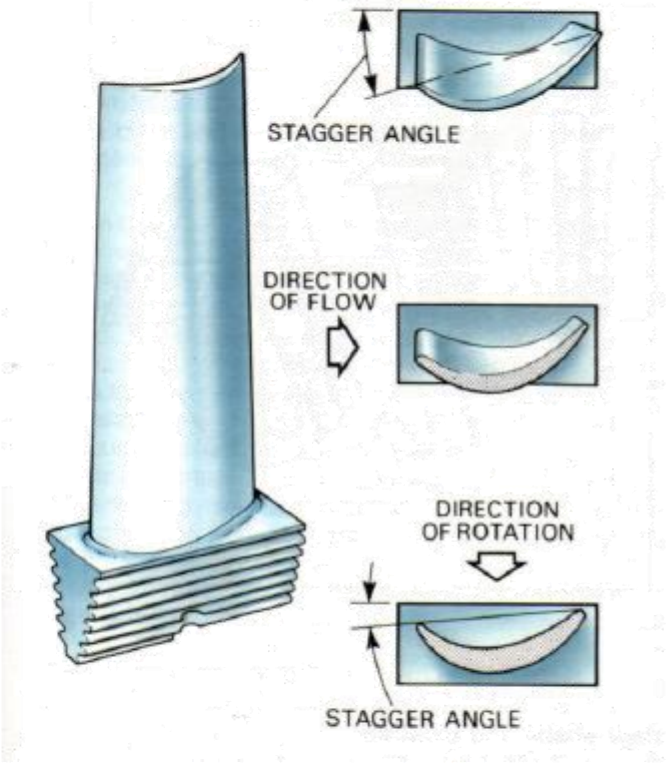




- Efficiency over most of operating range, constant.
- No change in flow function with speed once flow is choked.
- Most HP turbines operate in a choked condition.
- LP turbines will have more of their operations in the non choked region.

Turbine Layout





A typical uncooled turbine blade showing twisted contour

Film cooled HP nozzle guide vane

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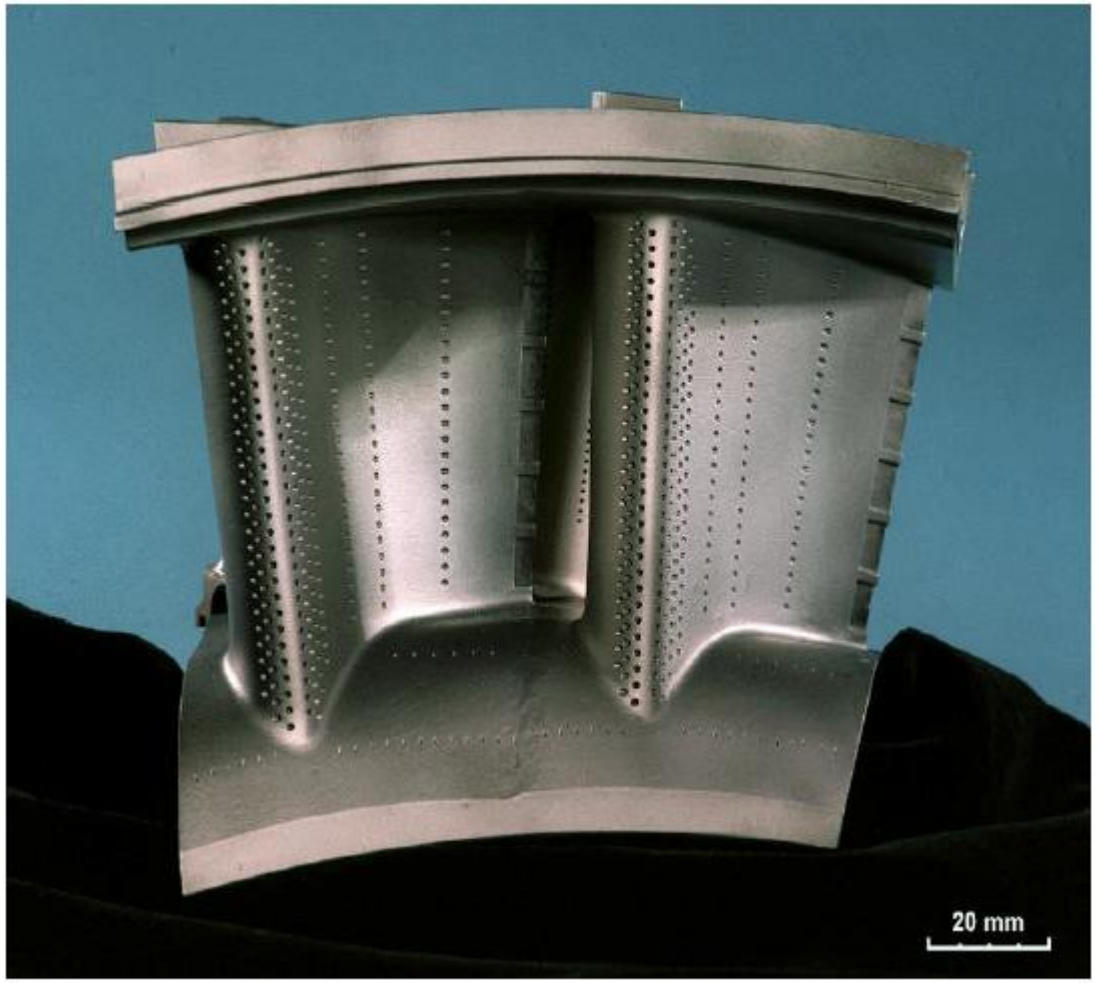
Turbine Materials

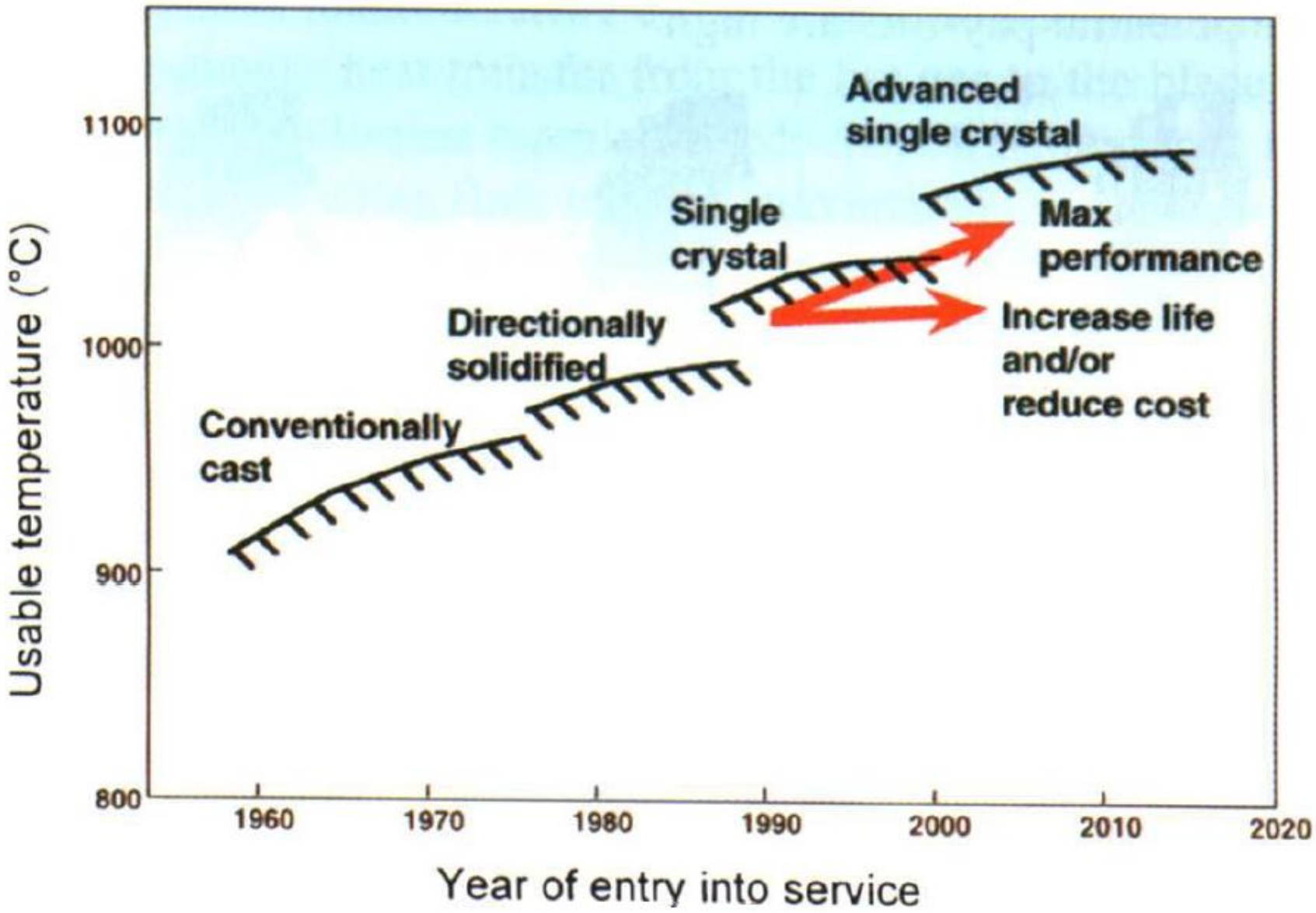
Nickel based Super-alloys

Typically: Nimonic 115

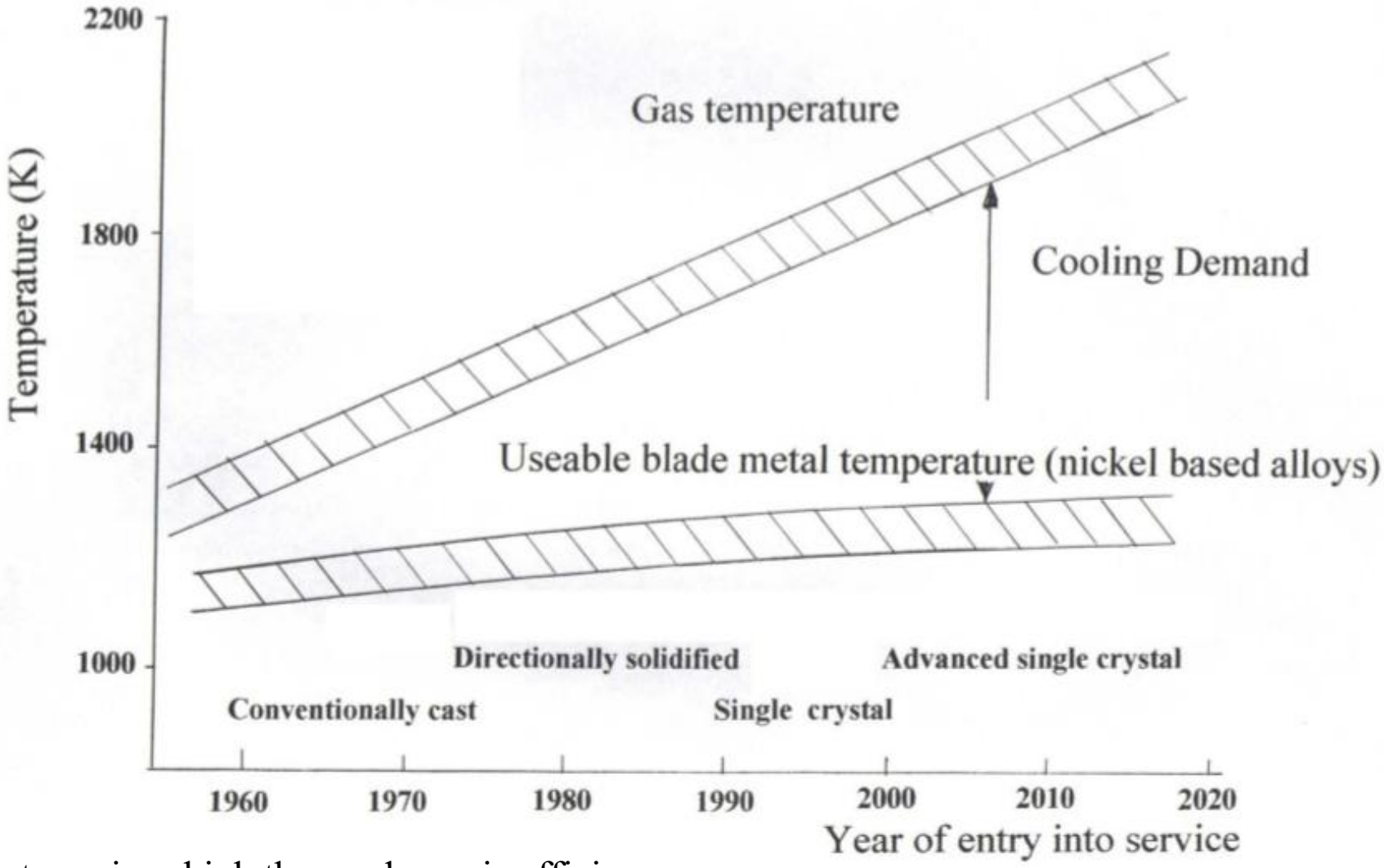
Nickel	57.3%
Cobalt	15%
Chromium	15%
Aluminium	5%
Titanium	4%
Molybdenum	3.5%
Carbon	0.16%
Zirconium	0.04%
Boron	0.014%



Turbine blade temperature capability



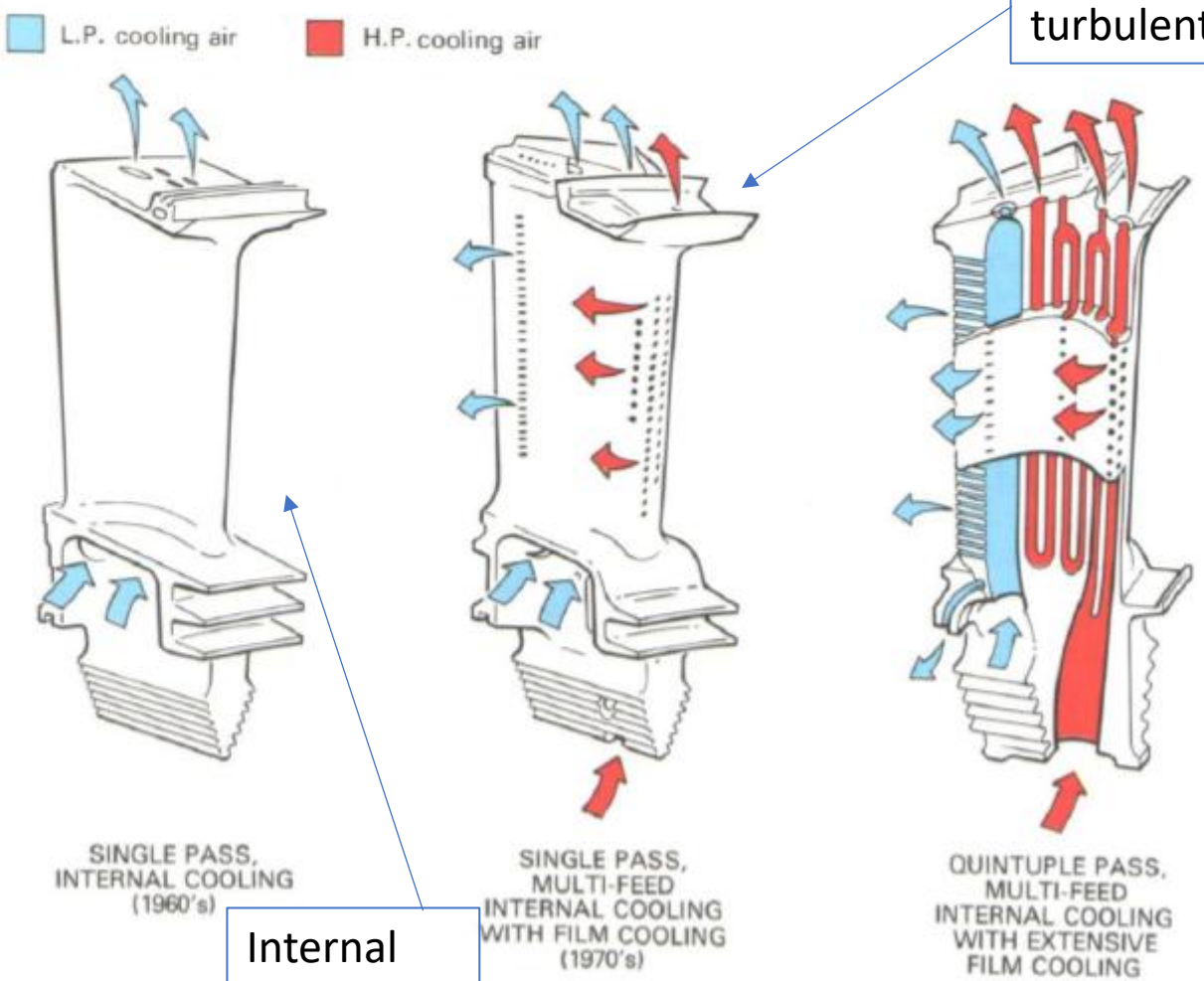
Turbine cooling



- High temperature gives high thermodynamic efficiency
- Blades must be cooled
- Balance between efficiency cost and life

The Development of HP Turbine Blade Cooling Systems

- Multi-pass convection and film cooling
- Cooling system designed to avoid 'hot spots'
- Must cool sufficiently to achieve blade life
- Requires knowledge of the heat transfer



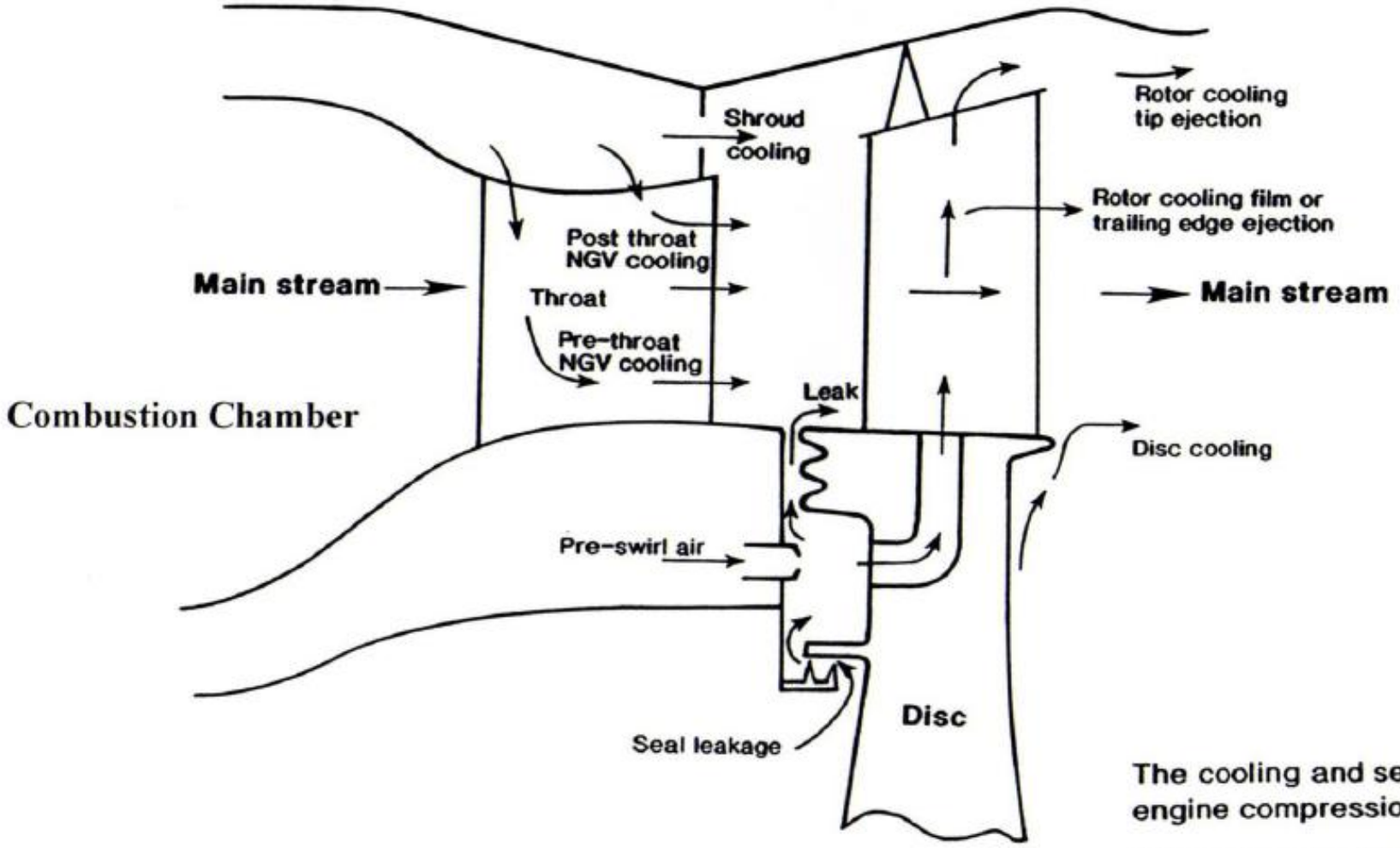
Film Cooling and turbulent mixing

Internal convection

Courtesy of Rolls Royce

Typical Turbine Cooling System

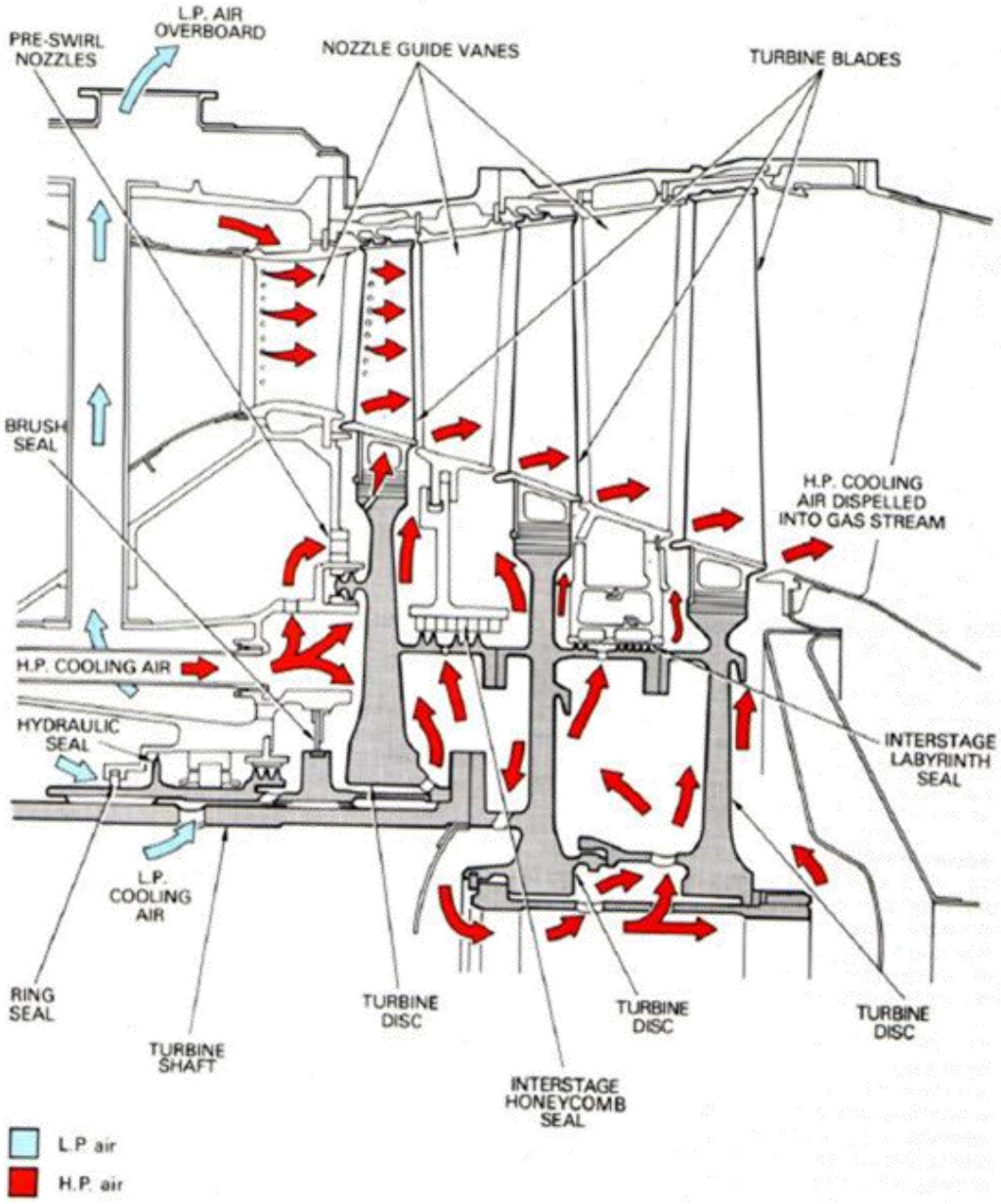
Schematic



The cooling and sealing air source is the engine compression system

There is a performance loss associated with the provision of cooling and sealing air.

Typical Turbine Cooling System



Turbine blade metal temperatures

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Cooling Effectiveness

$$= \frac{T_b - T_{cr}}{T_{g\,rel} - T_{cr}}$$

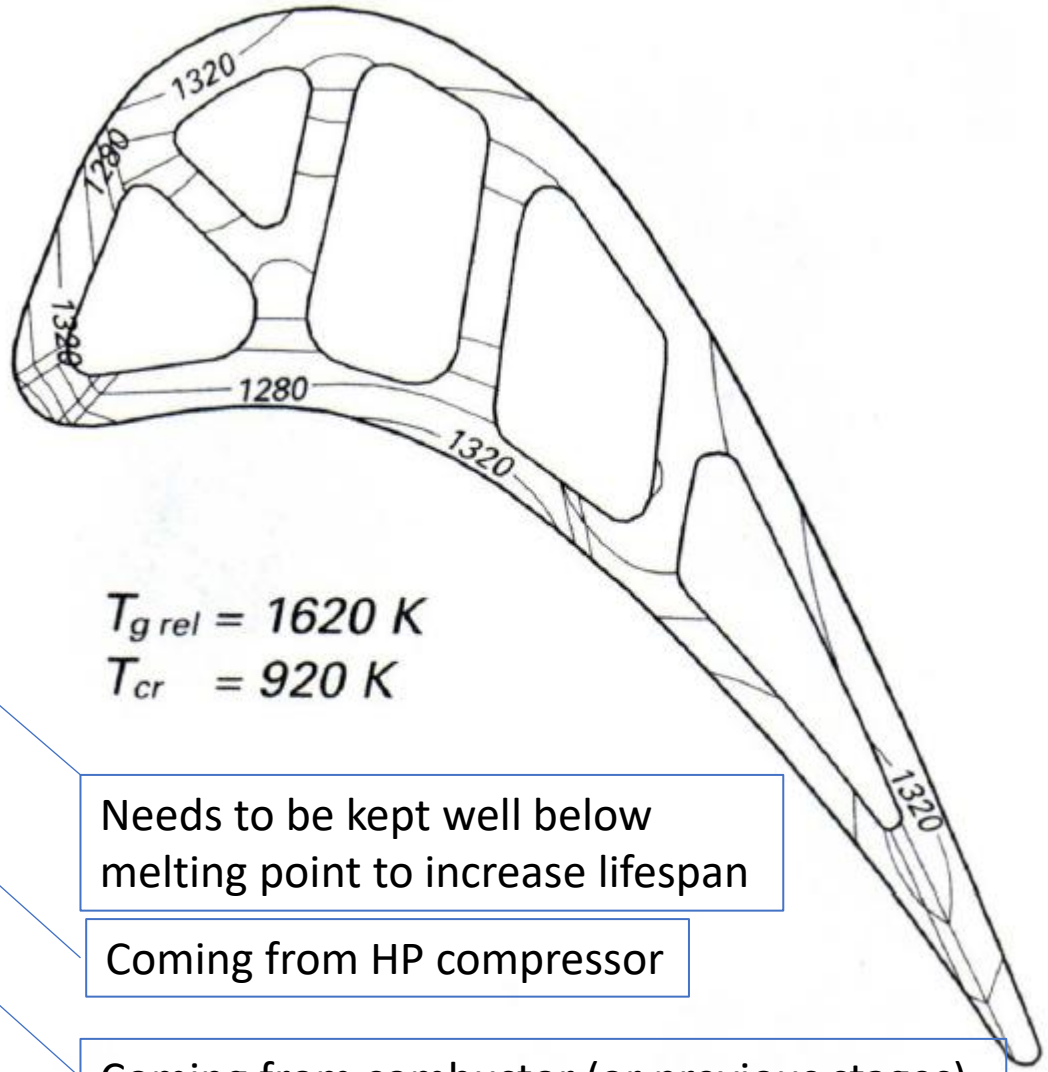
Where:

$\mathbf{T_b}$ = Mean blade metal Temperature

$\mathbf{T_{cr}}$ = Cooling Air Temperature

$\mathbf{T_{g\,rel}}$ = Mean Gas Temperature

Cooling airflow must be kept to a minimum



$T_{g\,rel}$ = 1620 K
T_{cr} = 920 K

Needs to be kept well below melting point to increase lifespan

Coming from HP compressor

Coming from combustor (or previous stages)

Turbine Blade Cooling

- In high performance engines, HP turbines NGV's & Blades are nearly always cooled.
- Intermediate Pressure (IP) (3 shaft layouts) and Low Pressure (LP) turbines maybe cooled depending upon cycle.
- The Designer will try & ensure as few stages as possible are cooled, as cooled turbine NGV's & blades are much more expensive than uncooled ones!
- For simple cycles and/or small gas turbines the cycle may not demand cooled blades

Turbine Blades

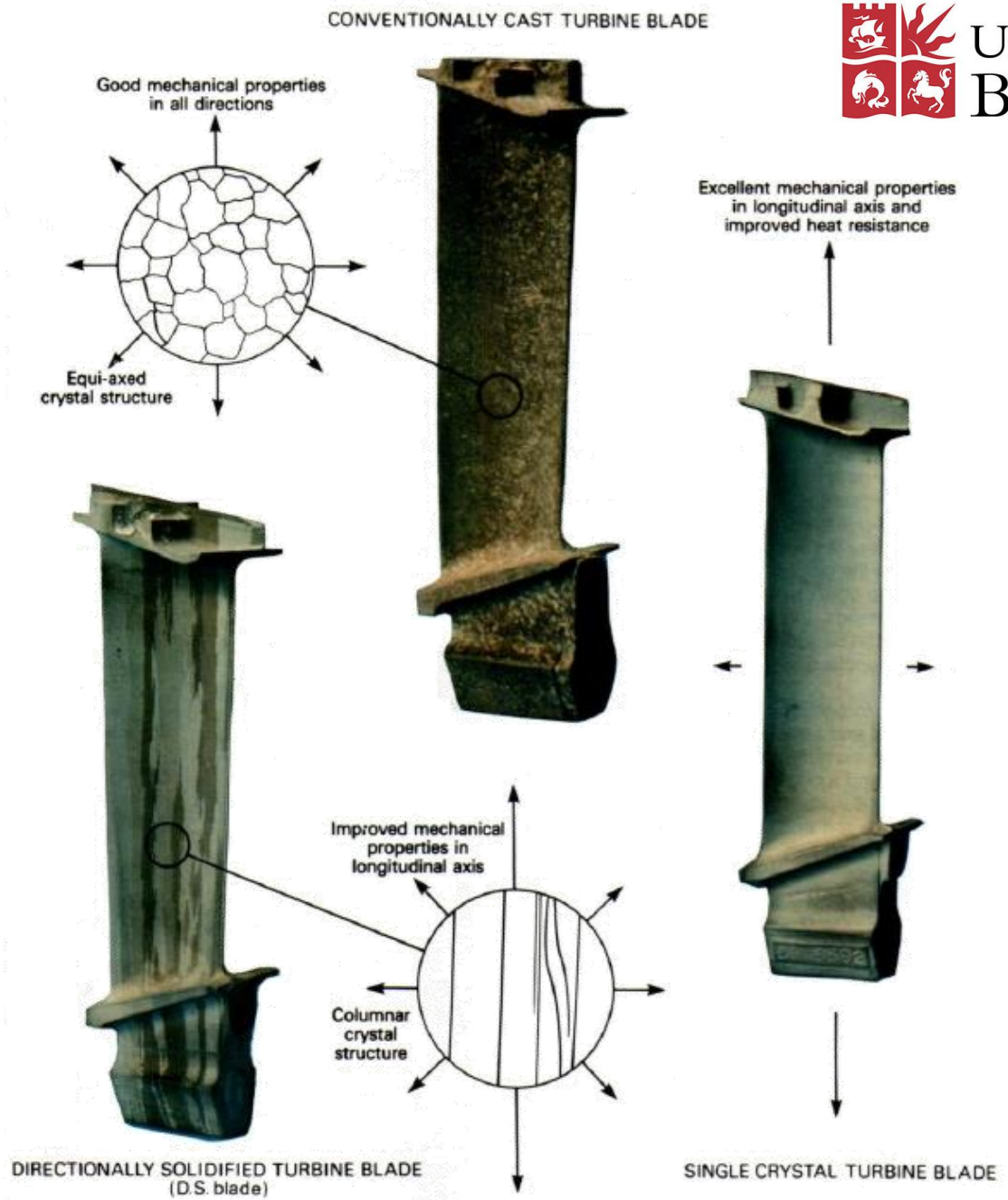
Conventionally cast

Directionally solidified

Single crystal

Video on the manufacturing process of Turbine Blades

https://www.youtube.com/watch?v=QtxVdC7pBQM



Key take-aways of Lecture 8

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- Fewer aerodynamics constraints with respect to compressors
- Materials & cooling key to high cycle efficiency
- Complex three dimensional, highly viscous flow
- Flow is inherently unsteady
- Gas temperatures up to 1600 deg C

What's on next week?

- Aircraft emissions and the environment
- Combustion processes